

West Bengal State Electricity Distribution Company Limited.
Office of the Chief Engineer, Procurement & Contracts Department,
Bidyut Bhavan (4th floor), Bidhannagar, Kolkata-700 091.

TECHNICAL SPECIFICATION FOR , 11 KV , 400 AMPS ROCKING TYPE TPGO , ISOLATOR

PART -A : Metallic Part

1. SCOPE :

This specification covers Design, Manufacture, Testing, and Supply of Three Post Rocking Type, Single Air Break 11 KV, 3 Phase, 400 Amps Gang Operated, Off-Load Isolator suitable for outdoor installation on DP / Sub-Stn Structures with either Horizontal or Vertical mounting.

2. APPLICABLE STANDARD:

The Isolator shall conform to prevailing specification IS-9921 (Part-I to IV) [1981 - 85] with latest amendment in general and to our specification in particular.

3. CLIMATIC CONDITION:

The atmosphere in the area is laden with industrial gases and smokes with dust in suspension during dry months and subject to long colder months. The temperature variation "between the daily average and maximum is large. Humidity occasionally rises up to 100% Heavy lightning is usual in this area during the month of May to November. The area is also subject to heavy monsoon rains. 80% to 90% of the annual precipitation being in the month of June to October.

DETAILED ATMOSPHERIC CONDITION IS STATED BELOW:

a)	Max. Tem. of air in sheds	50 deg C
b)	Min. Tem. of air in sheds	4 deg C
c)	Max. Temp. of air in Sun	60 deg C
d)	Max. daily average ambient temp	45 deg C
e)	Max. yearly average ambient temp.	30 deg C
f)	Max. relative humidity	100%
g)	Nos. thunderstorm days per annum.	100
h)	Average Nos. of dust storm days per annum	5
i)	Average No. of rainy days per annum	80
j)	Average no. of stormy rain fall days per annum(exceeding 4" in 24 hours)	10
k)	Average rainfall per annum	200 Cm
l)	Max. wind pressure	100 Kg /m ²
m)	Average wind pressure	71.3 Kg /m ²

4 GENERAL ARRANGEMENT & CONSTRUCTION :

4.1 The Isolator shall be of Outdoor & Channel-mounted type for Single Break Operation with Two Fixed Posts and one Rocking Post per Phase.

4.2 The Isolator shall be provided with Connectors at both ends.

4.3 Operating Mechanism with Padlocking Arrangement at 'OFF ' and 'ON ' Position :

The Operating Mechanism shall be fabricated from **M. S. Plate 38mm x 8mm x 305 mm in length**. The hinged socket which will take the vertical pipe should be made from 25 x 25 mm square and the Hinge arrangement should be perfect, so that the vertical up and down operation takes place without any obstruction.

The clamp of crank lever (Lever Angle) should also be made from 25 x 5 mm M.S. Flat with sufficient room to take the square shaft. The G.I. bolts should not be less than 10 mm dia for clamping the square bolt. Eye type intermediate guide should be of 12 mm dia 306 mm long or as necessary in length provided with nut, lock nut and 2 nos. flat washers.

Single piece 25 mm square M.S. Galvanized Phase coupling shaft shall be of length 2000 mm.

- i) Vertical operating **G.I. pipe 25 mm nominal bore with 12 SWG (2.65 mm) wall thickness 6 mtr in length in not more than two equal lengths with coupler shall have to be supplied .**
- ii) The handle for 'operating pipe' should be suitable for gripping by two hands of the operator and the same should be covered with corrugated polythene / rubber sleeve. 'ON' and 'OFF' indications shall be indelibly marked on weather proof and corrosion proof plates riveted to the body of the Isolator operating handle.

4.4. Fixed Contacts :

- i) The fixed contact shall be made from **Electrolytic copper strip 50 mm x 3 mm jaw type to be electro-silver plate with 8 micron deposit.**
- ii) The fixed Contact Guards shall also to be made from **M.S. Strip 50 mm x 1.5 mm (16 SWG) Shaped properly and galvanized.**
- iii) Two nos. Stainless steel springs made from 16 SWG wire shall be used at both side of the Fixed contact through 6 mm dia. 90 mm long stainless steel stud threaded at both ends and provided with stainless steel nuts check nuts and flat washer. The contact pressure of not less than 50 kg has to be achieved. The Spring Mechanism shall have to be provided so as to ensure that the speed of opening of contacts is independent of the speed of manual operation.

4.5 MOVING CONTACT :

The Moving contacts shall be made from **20 mm dia Hard-Drawn Electrolytic Copper Rod. One end of the Rod shall be flattened upto 175 mm in length** by Drop Forging arrangement and provided with hole so that the same is looked on the moving insulator. The other end is to be bent suitably for making contact into the Fixed Contact Jaw (forming Line-Contact). The palmed portion of the **copper rod should have minimum 28/30 mm width x 8 mm thickness.** The bolt used for fixing the Moving Contact with the Insulator Cap shall be of stainless steel. The round portion of the moving contact shall be silver electroplated with 8 micron deposit.

4.6 TERMINAL ARRANGEMENT :

Terminal plate shall be made from Hard Drawn **Electrolytic copper flat 38mm x 8 mm x 156 mm length** so that the Arcing horn and the Connector may be clamped. **Terminal pad** at the ongoing side (back plate) shall also be made from **electrolytic copper 38 mm x 8 mm x 140 mm.** A **Pantograph holder** made of **Galvd. 38 mm x 8 mm having an eye at the centre** to take one end of the pantograph assembly shall be fitted with the terminal pad on the insulator top.

4.7 FLEXIBLE BREADED TAPE :

A flexible link made from fine tinned copper braided tape of **minimum size 40 mm (width) x 6 mm thickness having minimum weight of 0.900 kg / meter** length shall connect the Moving contact with the Terminal pad through the Pantograph. The flexible tape will be treated as a part of the isolator and the temperature rise should be within the allowable limit as per ISS when full load current is passed through the isolator. **The length of the braided tape will be 1000 mm in each phase** both ends of the tape shall be suitably terminated by Tinned Copper sockets securely fixed to the Terminal pad of the Rocking Insulator with nuts, bolts and washers etc, at one end and the other end at the other end at the outgoing Terminal I-pad. The connectors shall be such as to facilitate easy replacement during future maintenance.

4.8 PANTOGRAPH ASSEMBLY :

The **Pantograph** shall be fabricated from **30 x 3 mm M.S. Flat** duly galvanized and fitted with hinge for folding and unfolding the Pantograph during 'Back and Forth' movement of the insulator assembly. It shall have efficient width to accommodate the flexible tape Twisted M.S. Flat 25 x 3 mm pieces shall be used to hinge the Pantograph ends with Rocking & Fixed Insulators.

4.9 FIXED AND MOVING ARCING HORN:

Phosphor Bronze Arcing Horn are to be made from **8mm dia and smooth** and even surface. The two parallel limbs of the Fixed Arching Horn 'shall be spaced strictly as per drawing so that, the spring action remains unaltered and permanent. The Moving Arcing Horn' will have 'Eye' at one end for bolting with the Moving Contact at its flat space and in addition, a clamp made of non-ferrous strip 16mm wide, 20 SWG thick to be provided for tightening the Arcing Horn with the Moving Contact at round portion. Feature of the Arcing Horns should be to Make Before & Break After main contacts so that they can break the magnetizing/ changing current to avoid any damage to the contacts.

4.10 CONNECTORS :

Single grooved bolted type connector to take the ACSR 'DOG' conductor shall be fitted with **copper - Aluminium Bi-metal of size 50 x 38 x 10 mm** at the incoming (Fixed Copper) side and other Connector shall be fitted at the outgoing end of the 'Flexible Jumper' The palm of the **Connector should be of size 50 x 38 x 10 mm tapered 50 mm in length**. The composition of Aluminium alloy shall conform to the following composition:-

a)	Aluminium	80% minimum
b)	Iron	Less than 1%
c)	Silicon	Less than 15%
d)	Cu-Mg. Zn Pb	Less than 0.25%
e)	Manganese	Less than 1.0 %

a. ROCKING BASE :

The Moving Post Insulator shall have to fitted on a Rocket to 'Back' and 'Forth' movement. M. S. fabricated Roller Body shall be supported on m.s. Fulcrums fitted at both side of the Base Channel with bolts & nuts. The side supports shall have Brass Bush / Glass-Nylon bush bearing for easy swivel movement of the Roller Body upon which the Insulator column is fitted. The design of the roller body (swival) shall be such as to make it capable of withstanding mechanical endurance test as per relevant IS.

b. BASE CHANNEL :

The **Base Channel shall be of size 75 mm x 40 mm x 5 mm of suitable length without any sharp edge**. The channel should be hot rolled quality.

4.11 EARTHING :

- The 11KV system is effectively earthed. The Isolator shall be provided with Earthing Terminals, fixed with bolts and nuts, the diameter of the bolts being 12 mm in at least one of the three Base Channels provided in each set of Isolator.
- Each Operating Handle shall be provided with Earthing terminal along with Bolt and Nut of above mentioned sizes.
- All the Earthing Terminals shall be marked invariably with Earthing symbol.

4.12 PROTECTIVE COATING :

All ferrous parts must be Hot Dip Galvanized as per IS-2633 1972 or latest amendments thereof and must successfully withstand four successive one minute dip in CuSo4 solution of specific gravity 1.186 as prescribed in relevant ISS.

4. MARKING :

Name Plate shall have to be provided in each channel of Isolator with the information as specified in clause no. 10.1 of 15:1818-1972. The Name Plate - should be weather proof & corrosion proof. Manufacturer's serial number and customer's order reference shall be indicated in the 'Name Plate'

5 TOLERANCE:

+/- 5% Allowance shall be allowed in the measurement of dimensions for our acceptance

6 PACKING :

Materials Shall be packed by wooden crates of suitable thickness to protect against damage in transit and in way that individual unit can be dispatched at a time complete in all respects so that erection is not held up for want of any components.

7 DRAWING :

The complete isolator shall be manufactured and supplied against this specification strictly in accordance with the following drawings enclosed with the Specification.

DRAWING NOS:-

- | | | |
|----------------------|----------------------|----------------------|
| 1. S&P/MC/II KV/I/1 | 2. S& P/MC/II KV/I/2 | 3. S& P/MC/II KV/I/3 |
| 4. S& P/MC/II KV/I/4 | 5. S& P/MC/II KV/I/5 | 6. S& P/MC/II KV/I/6 |

However, GA drawings of Isolators including different components shall have to be submitted by the Tenderer with the Bid.

After placement of order, drawings are to be submitted & approved before making offer for inspection & testing.

8 ELECTRICAL AND MECHANICAL CHARECTERIST1CS:

i)	Nominal system voltage	11 KV RMS
ii)	Highest system voltage	12 KV RMS
iii)	System frequency	50 Hz ($\pm 3\%$)
iv)	Maximum continuous current rating	400 Amps.
v)	Rated short time current rating for 1 Sec.	18.4 KA
vi)	No. of poles	3
vii)	Phase center for Isolators	750 mm
viii)	Min. clearance between fixed and moving contacts	310 mm
ix)	Height of the post Isolators to be used in the Isolator.	254 mm
x)	Centre to Centre distance of mounting holes at the base Channel	480 mm

In 'Closed' Position, the Isolating switch must withstand the mechanical and thermal effects of Short Circuit Current for 1 Sec.

9. COMPLETENESS OF SUPPLY

Any fittings, Accessories or Apparatus which may have not been mentioned in this specification but which are necessary for Isolators' efficient operation, performance and satisfactory maintenance shall be deemed to be included in the Contract and shall be supplied by the Contractor without any extra charges. All Insulators shall be complete in all respects whether the details are mentioned in the specification or not. Nuts which may work loose in operation must be provided with spring washers / split pins.

NOTE

The Technical specification contained in this section are for guidance of the Tenderer. Deviation if any, from the Purchaser's specification proposed by the Tenderer may be considered provided those are necessary either to improve the Utility, performance and Efficiency of the Equipment to secure overall economy. GA Drawings of Isolators including different components shall have to be submitted by the Tenderer with the Bid showing details of equipment.

The complete isolator shall be manufactured and Supplied against this specification except that for supporting which are excluded from the scope of this specification.

The supporting structure for the isolators are excluded from the scope of this specification.

Drawings for the details of structure for mounting of the isolators shall be furnished by the Purchaser to facilitate design to mounting arrangement etc., to the concerned controlling officer prior to actual execution.

Pedestal Insulators shall be supplied as per specific requirements laid down in Part-B of this Technical specification.

Test certificates for the offered Isolators including Insulators shall be furnished as per Guide lines laid down in Part-C of this Technical Specification.

G.T.P. and other informations of the offered Isolators shall be furnished as per proforma given in Part-D of this Technical specification.

Technical particulars of offered Insulators shall be furnished as per proforma given in Part-E of this Technical specification.

Tenderer shall furnish document including Test certificates and Performance certificates in support of supply & delivery of similar equipment indicating thereon the names of the organization, quantity ordered, quantity supplied along with the tender.

Manuals for installation operation and maintenance procedures shall have to be submitted (in six copies) along with the design drawings.

The price (quoted price) should include the price of equipment including all accessories as per schedule of requirement.

While quoting, the supplier shall furnish the recommended list of spares required for a period of three years of operation, item wise prices for the spares shall be furnished separately.

Bill of materials shall be noted in the G.A. drawing as per following Format.

MARK	DESCRIPTION	QUANTITY	MATERIAL	REF.DRG.NO.

PART-B

(A) SPECIFIC REQUIREMENT OF PEDESTAL INSULATOR (FOR 11KV 400 A TYPE)

- Each set of 11 KV. TPGO Isolator shall be provided with 9 Nos. of 11KV Pedestal insulator duly fitted .
- The Post Insulator shall in general conform to the latest issue of IS: 2544/1973 with all amendments thereof.
- The porcelain shall be sound , free from defects , thoroughly verified and smoothly glazed . **The Post insulator shall be Brown in colour** as per general practice. The glaze shall cover all the porcelain parts of the insulator except those areas which serve as support during firing or left unglazed, or the purpose of assembly.
 - Cement used in the construction of insulators shall not cause fracture by expansion or loosening by contraction and proper care must be taken in 'curing' cement shall not give rise to chemical reaction with the metal fittings and its thickness shall be as uniform as possible.
 - The design of the insulators shall be such that stress due to expansion or contraction of any part of insulators shall not lead to deterioration. The porcelain shall not engage directly with hard metal.
 - The post insulators should be cemented by manufacturer of insulator. Suitability of cementing shall be ensured by required tests specified in the relevant Indian Standards.

(B) SPECIFIC REQUIREMENT OF POST INSULATOR (FOR 11 KV,400A TYPE)

1. The post insulators shall be cemented only by the manufacturer of insulator. Suitability of cementing shall be ensured by required tests specified in the relevant IS :
2. At the time of inspection, one or more insulator(s) may be dismantled by the Purchaser's Representative from the isolators offered for inspection and the contractor shall arrange at his own cost for both Mechanical and Electrical Tests at the insulator manufacturer's laboratory within reasonable time as per latest ISS failing which entire lot may be rejected.
3. The isolator supplier shall collect a certificate from the manufacturer of insulators that cementing has been done by them.

4. **SPECIFIC TECHNICAL PARAMETERS OF 11KV POST INSULATORS :**

Sl. No.	Description		11 KV Post Insulator
1	Nominal system voltage/Working voltage	:	11 KV
2	Highest system voltage	:	12 KV
3	No. of units per Stack	:	1
4	Minimum specific Creepage distance	:	25 mm per KV
5	P.F Visible Discharge Voltage	:	9 KV rms
6	P.F. Withstand Voltage		
	a) Dry	:	60 KV rms
	b) Wet	:	35 KV rms
7	Impulse withstand voltage 1.2/50 micro second wave :		
	a) Positive	:	75 KVp
	b) Negative	:	80 KVp
8	P.F. Minimum flash over voltage		
	a) Dry	:	70 KV rms
	b) Wet	:	45 KV rms
9	Impulse flashover voltage 1.2/50 micro second wave :		
	a) Positive	:	95 KVp
	b) Negative	:	120 KVp
10	Power Frequency puncture withstand voltage on single unit	:	105 KV rms
11	Cantilever strength on single unit		
	a) Upright	:	5 KN
	b) Inverted	:	3 KN
12	Tensile Strength	:	15 KN
13	Torsional Strength	:	200 Nm
14	Compression Strength	:	30 KN
15	Height	:	254 mm

Sl. No.	Description		11 KV Post Insulator
16	Insulation Part Diameter	:	152 mm
17	Pitch Circle Diameter		
	a) Top	:	57 mm
	b) Bottom	:	57 mm
18	Conforming standard	:	As per IS

PART C :

TEST CERTIFICATES / INSPECTION AND TESTING

TEST CERTIFICATES/INSPECTION AND TESTING

1. I S O L A T O R :

The bidder shall submit following copies of complete type test certificates of similar Isolator of identical rating & design as well as on 11 KV Post insulators already conducted in CPRI/NABL accredited/Govt. approved test house or laboratory containing NABL logo. Type test report shall be carried out within five years from the due date of submission of tender. The bidder shall have to submit the type test report along with the tender, otherwise the offer will be rejected.

Following type test report shall have to be submitted:

- a) Short Time Withstand & Peak Withstand Current Test.
- b) Lightning Impulse Voltage Withstand Test.
- c) PF Withstand Voltage Test (Dry & Wet)
- d) Mechanical Endurance Test
- e) Temperature Rise Test

In addition to that, the bidder have to submit Type Test Report on 11 KV Post Insulator as per relevant IS/IEC and technical specification of the Tender along with the Tender document.

The dimensions of the Fixed & Moving Contacts of the sample Isolators, tested successfully for Type Tests in an approved Laboratory, should at least conform to the dimensions as specified in Part – A of this specification. The type of Fixed Contact should be of Reverse Loop Type.

Following Acceptance Tests shall be carried out at the works of the Manufacturer as per relevant ISS before delivery of each lot in presence of our Engineers. The contractors shall furnish 'Routine Test' reports of each unit comprising 'Millivolt Drop Test', 'Operation Test' and 'One Minute Power Frequency Voltage Withstand Test' in six copies along with each lot offered :-

- a) Temperature Rise Test
- b) Millivolt Drop Test
- c) One minute power frequency voltage withstand test
- d) Operation Test : The tenderer shall make arrangement for operation test for the total height of the Isolators on selected samples.
- e) Uniformity of silver coating on the copper contacts
- f) Dimensional Check-up as per drawing and specification on selected samples.
- g) Mechanical Test for Insulator as per IS-2544
- f) Uniformity of zinc coating on ferrous parts (galvanization test)

This test shall be carried out at the Laboratories undertaking tests for Govt. Deptts. And in such case, reports incorporating Test Results from the respective Laboratories should be submitted in support of the contractor's Test Certificates .

Samples at random will be selected from the lot offered for above testing and if any one of the test piece fails, the lot will be rejected.

Contact pressure, Material Quality, Dimensions of the Isolator Contacts and other components shall also be verified /measured (with the approved Drawing). Quoted price should include charges for above stated Acceptance Tests and Type Tests.

Testing of materials: The purchaser reserves the right to get the metal/raw materials/silver coating/galvanizing of any component, tested at any approved laboratory at the cost of manufacturer, in case of any doubt over their quality of such metal / raw material especially for Copper and Aluminium components.

PEDESTAL INSULATOR :

1. The Pedestal Insulator to be fixed and supplied along with the Isolators will be tested both for Mechanical and Electrical characteristics of the complete insulator as per late issue of ISS - 2544 / 1973. Complete Type Test Certificates from the Manufacturer shall also to be furnished, for both Electrical and Mechanical Tests.
2. One or more Insulator (s) may be dismantled by the Purchaser's Representative, from Isolators offered for inspection and the Contractor shall arrange at his own cost for both Mechanical and Electrical Tests at the insulator manufacturer's Laboratory within reasonable time as per latest ISS.
3. The Test values in respect of 'Impulse Voltage Test' and 'Power Frequency Voltage Withstand Test' on the complete isolators without grounding connectors, should not be less than those specified in ISS and as detailed below :

System Voltage	Impulse Withstand Voltage Positive & Negative Polarity		One Minute Power Frequency Withstand Voltage (RMS)	
	Across Isolating Distance	To Earth and Between Poles	Across Isolating Distance	To Earth & Between Poles
11 KV	85 KV	75 KV	32 KV	28 KV

Rated short time current rating is 18.4 KA (rms) for 1 Sec.

3. The equipment shall be subjected to routine & acceptance tests in accordance with provisions of relevant standards.

Routine & Acceptance tests shall have to be performed in presence of purchaser without any extra cost. The supplier/ manufacturer shall give at least 15(Fifteen) days" advance notice for conducting such tests.

Purchaser shall have the right to select any number on random sampling basis for testing from the equipment ordered for inspection and in the event of failure in such test, the purchaser shall have the right to reject the whole lot.

Six (6) copies of routine & acceptance reports shall have to be furnished to the purchaser for approval before dispatch of the offered equipment.

4. TYPE TESTS AFTER ISSUANCE OF ORDER:

Besides submission of Type Test Report, carried out within five years as per Tender Specification, Type Test at the discretion of Ordering authority, shall have to be arranged by the successful contractor or WBSIEDCL on his own may conduct the same from any lot offered for inspection, with sample chosen at random after

successful routine test by our Inspection team, as per the relevant ISS from any NABL accredited/ Government recognized Test House or Laboratory in presence of WBSEDCL'S representative. In case of failure of the materials after type test, WBSEDCL will have the right to reject the total supplied lot of the said materials and the party have to replace the complete lot of materials at his own cost including transportation of materials at site.

However, the necessary cost of the type test charges will be reimbursed to the party on production of necessary supporting documents.

GUARANTEE:

In the event of any defect in the equipment arising out of faulty design, materials, workmanship within a period of **5 (five) years from the date of last despatch** of any integral part of the equipment/cable, the supplier shall guarantee to replace or repair the same to the satisfaction of the purchaser.

If the supplier fails to do so within one month of receipt of intimation, WBSEDCL reserves the right to effect repair or replacement by any other agency and recover charges for repair or replacement from the supplier.

4. DOCUMENTS TO BE SUBMITTED AT THE TIME OF PHYSICAL DELIVERY TO THE CONSIGNEE STORES

The following documents to be submitted by the Vendors to the Consignee Stores at the time of physical delivery :-

- a) Copy of Purchase Order.
- b) Copy of Despatch Instruction.
- c) Inspection Test Certificate.
- d) Guarantee Certificate.
- e) Proforma Invoice.
- f) Seal list and packing list.
- g) Challan in triplicate.
- h) Way bill, if applicable.

ENCLO: ANNEXURE D & E

**SCHEDULE OF GUARANTEED TECHNICAL PARTICULARS FOR 11KV 400A ROCKING TYPE TRIPLE POLE
GANG OPERATED AIR BREAK ISOLATING SWITCHES**

(TO BE FILLED BY TENDERERS)

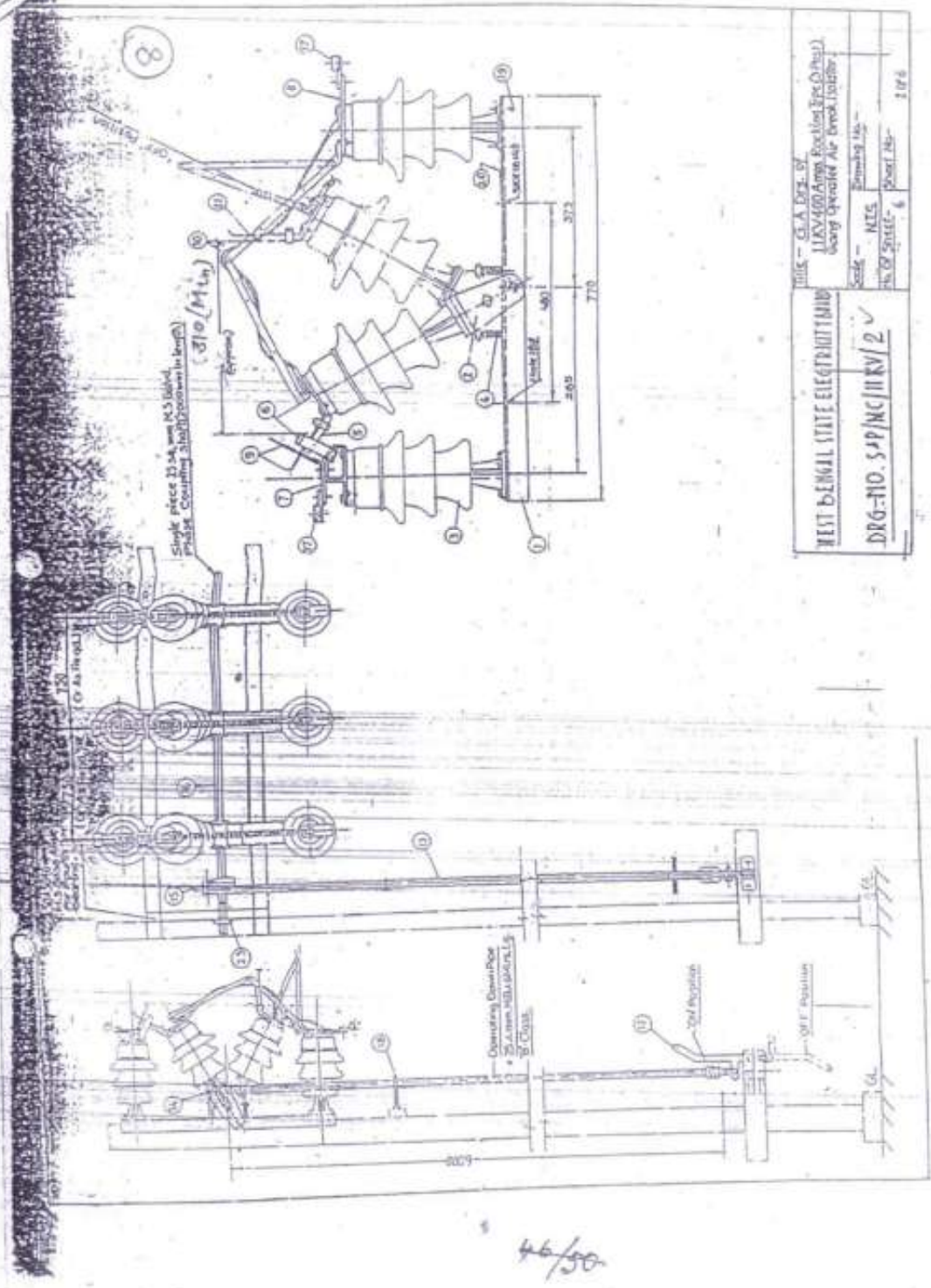
1.	Name of manufacturer of Isolating Switches	
2.	Particulars of Isolating Switches:	
a)	Number of Poles	
b)	Voltage rating	
c)	Current rating in Amps :-	
	i. Normal(Continuous rating)	
	ii. Maximum & Short time current rating for 1 Sec.	
d)	Type, Material and contact pressure (in kg.) of Cu.contacts and stainless steel spring :-	
	i. Moving Contact	
	ii. Fixed Contact	
	iii.Compression Spring	
e)	Whether contacts are silver faced or tin plated(the thickness may be indicated)	
f)	Temperature rise obtained during type test as per clause No.11.1.7 of IS:1818 of the following at the full rated current in degree centigrade over an ambient temperature of 50 °C	
	i. Contacts	
	ii. Terminals	
	iii. Compression Springs	
g)	Measurement of the resistance of the main circuits(in milli volt drop) which was obtained during type test as specified in relevant Clause of IS :1818 and IS 9921	
	i. Contacts(fixed and Moving)	
	ii. Terminals	
h)	Frequency	
i)	Type and material of the terminal connector provided	
j)	Diameter, wall thickness and length of G.I.Operating pipe	
k)	Size and length of base mounting M.S. Galvanised channel provided for the three phase :	
l)	Minimum clearance between phases in mm. and the centre distance between the Insulators of the adjacent phases in mm. in the assembled position of the switch :	
m)	Mounting /Fixing Hole Centre Distance in the Base Channel and Sizes	

Signature

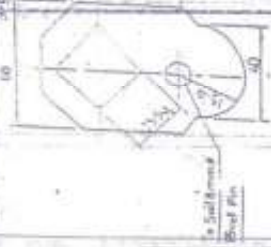
SCHEDULE OF TECHNICAL PARTICULARS FOR 11KV 400 A PEDESTAL INSULATORS**(TO BE FILLED BY TENDERERS)**

SL.NO.	<u>ITEM</u>	
1.	<u>Particulars of Insulators:</u>	
	a) Type of Insulators	
	b) Name of Manufacturer of Insulators.	
	c) Height of the Insulators	
	d)Diameter of the largest Shed	
	e)No of units per stack	
2	<u>ELECTRICAL CHARECTERISTICS (FOR ONE INSULATORS)</u>	
2.1	<u>FLASH OVER VOLTAGE</u>	
i)	Dry Power frequency	
ii)	Wet Power frequency	
iii)	Impulse voltage of 1.2/50 micro sec.(+ve)	
iv)	Impulse voltage of 1.2/50 micro sec.(-ve) :	
2.2	<u>WITHSTAND VOLTAGE</u>	
i)	Dry voltage	
ii)	Wet voltage	
iii)	Impulse voltage of 1.2/50 micre sec.(+ve)	
iv)	Impulse voltage of 1.2/50 micro sec.(-ve)	
2.3	<u>VISIBLE DISCHARGE VOLTAGE(KVrms)</u>	
2.4	<u>POWER FREQUENCY PUNCTURE WITH STAND VOLTAGE OF UNIT</u>	
2.5	<u>MECHANICAL CHARACTERISTIC</u>	
a)	Cantilever strength upright	
b)	Cantilever strength under hung	
c)	Torsional strength	
d)	Tensile strength	
e)	Compression strength	
2.6	<u>GENERAL CHARACTERISTIC</u>	
a)	Minimum Creepage distance	
b)	Weight of complete unit	
c)	Weight of non ferrous Parts(total)	
d)	Weight of ferrous Parts(total)	
2.7	<u>STANDARD TO WHICH INSULATOR CONFORMS</u>	

Signature



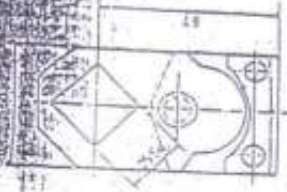
WEST BENGAL STATE ELECTRICITY BOARD DRG-NO. SP/NC/11KV/2	
TITLE - C.A. DRG. OF 11KV/400V Air Break Isolator (Type Open) Using Operated Air Break Isolator	
Scale - N.T.S.	Drawing No. -
No. of Sheet - 6	Sheet No. - 2 of 6



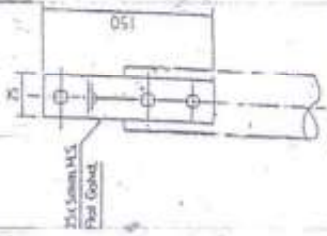
UPPER PART OF
FOURTH BEARING



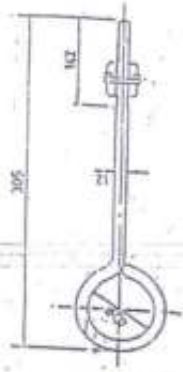
LOWER PART OF
FOURTH BEARING



FOURTH BEARING ASSY.



TOP PIPE CLEVIS

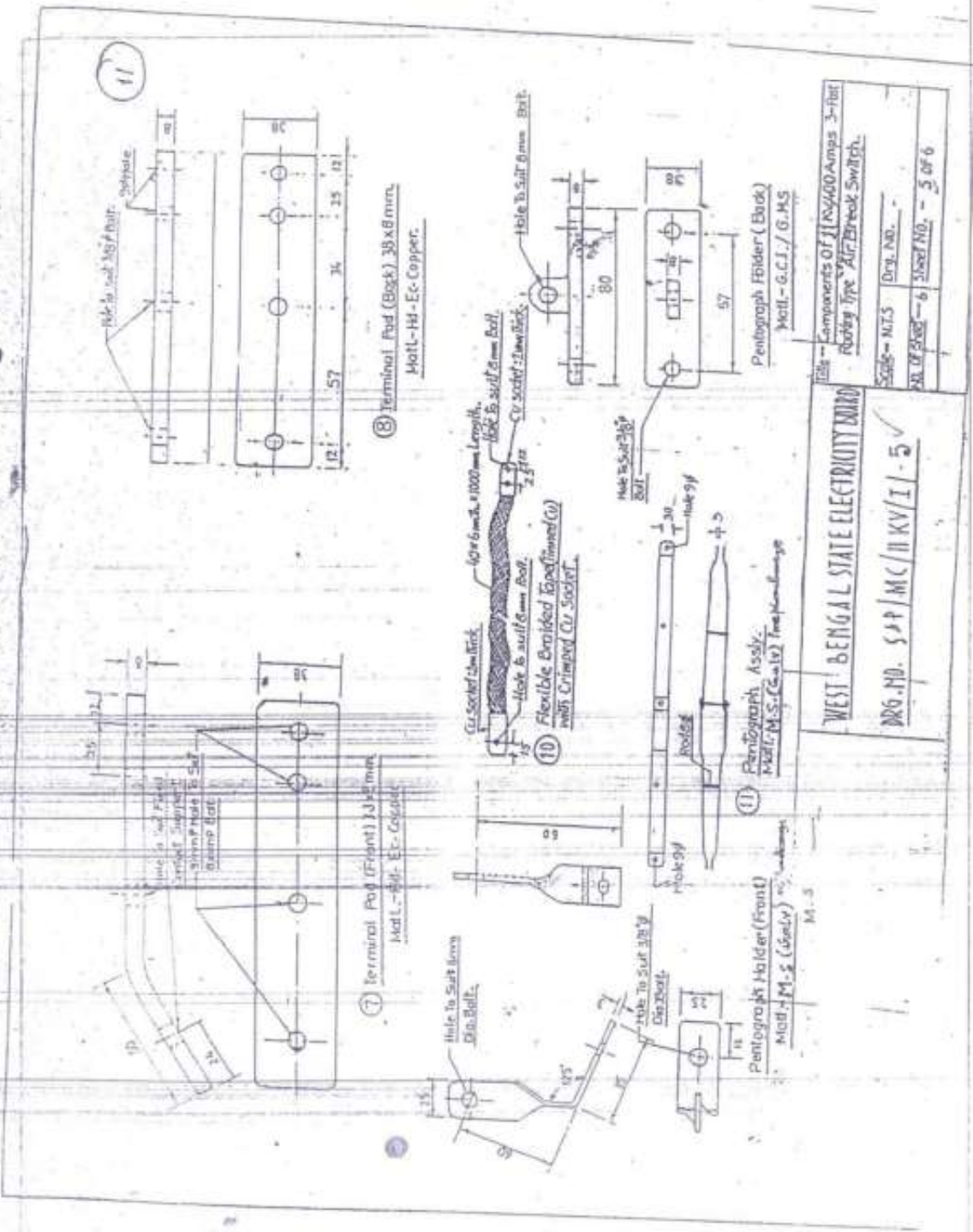


INTERMEDIATE GUIDE
(Eye Type)

WEST-BERKSHIRE CITY BOARD
COMPONENTS FOR 11KV 400A ROUGH
GANG OPERATED AIR BREAK ISOLATOR
SCALE: N.T.S.
ORG. NO. 51/MCH/11/14
DATE: 10/1/50
BY: [Signature]
CHECKED: 4 OF 6

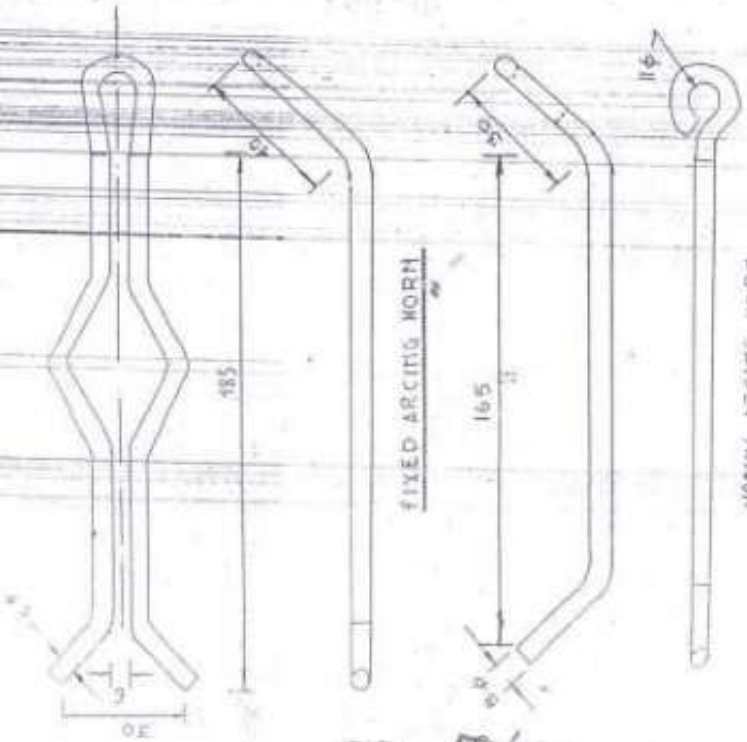
48/50

51



49/50 52

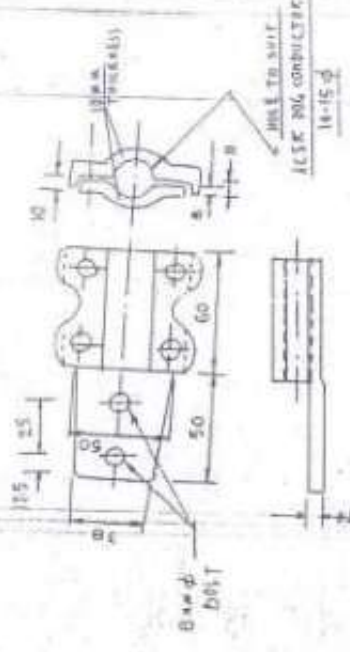
12



FIXED ARCING HORN

MOUNTING ARCING HORN

ALL DIMENSIONS ARE IN MM
MIL PHOSPHOROUS BRONZE



CONNECTOR

MIL - ALU. ALLOY

WEST BENGAL STATE ELECTRICITY BOARD	FILE NO. 507 MC/1012/6
DATE: 10/10/12	BY: 10/10/12
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